

# PAPER\_781: M74 Phantom Galaxy — UQFF Grand Design Spiral Reference

**Author:** Daniel T. Murphy

**Framework:** UQFF (Universal Quantum Field Superconductive Framework)

**Session:** 181 | v5.42

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**CP4 Class:** #365 — M74PhantomGalaxyUQFFCalculator

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## Abstract

M74 (NGC 628) is the “Phantom Galaxy,” a nearly face-on grand-design spiral approximately 32 million light-years away ( $z \approx 0.0022$ ) in Pisces. Known as one of the most symmetric spirals in the sky, M74 has been extensively studied by Hubble, Spitzer, and most recently JWST (which released stunning mid-infrared MIRI images in 2022 showing its spiral arms in extraordinary detail). With a moderate SFR  $\approx 1\text{--}2\text{ M}\odot/\text{yr}$  and total mass  $\sim 10^{11}\text{ M}\odot$ , M74 is a textbook Sbc spiral. Under UQFF, standard disk rotation ( $v = 10^5\text{ m/s}$ ) with typical galactic B-field yield  $g_{\text{M74}} \approx 1.053 \times 10^{-3}\text{ m/s}^2$ .

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## 1. Introduction

M74’s nearly perfect face-on orientation (inclination  $\sim 5^\circ$ ) makes it an ideal case for tracing spiral structure and measuring the undisturbed rotation curve. JWST MIRI imaging in 2022 revealed star-forming clumps, dust filaments, and HII regions threading the symmetric spiral arms at previously unresolved spatial scales. The symmetric spiral structure indicates a stable, long-lived disk without major recent mergers. UQFF captures M74’s regular disk rotation through standard  $v = 10^5\text{ m/s}$  coupling, with moderate  $M_{\text{sf}} = 0.045$  reflecting its  $\sim 1.5\text{ M}\odot/\text{yr}$  SFR averaged over 5 Gyr of evolution.

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## 2. Master UQFF Gravity Equation

$$g_{\text{M74}}(r, t) = (G \times M) / r^2 \times (1 + H(z) \times t) \times (1 + M_{\text{sf}}) \times (1 + f_{\text{TRZ}}) + a_{\text{EM}}$$

### 2.1 Parameters

| Parameter        | Symbol | Value                                                     | Source            |
|------------------|--------|-----------------------------------------------------------|-------------------|
| Galaxy mass      | M      | $10^{11}\text{ M}\odot = 1.989 \times 10^{41}\text{ kg}$  | Hubble/JWST       |
| Disk radius      | r      | $5 \times 10^{20}\text{ m}$ ( $\sim 53\text{ kly}$ )      | NED               |
| SFR              | —      | $1.5\text{ M}\odot/\text{yr}$                             | JWST MIRI         |
| Age              | t      | $5 \times 10^9\text{ yr} = 1.578 \times 10^{17}\text{ s}$ | Hubble time       |
| $M_{\text{sf}}$  | —      | 0.045                                                     | UQFF SFR integral |
| Redshift         | z      | 0.0022                                                    | Spectroscopic     |
| $v_{\text{EM}}$  | v      | $10^5\text{ m/s}$                                         | Disk rotation     |
| $B_{\text{EM}}$  | B      | $10^{-5}\text{ T}$                                        | Galactic field    |
| $f_{\text{TRZ}}$ | —      | 0.04                                                      | UQFF              |

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### 3. Long-Form Derivation

#### Step 1: Base Gravitational Term

$$g_{\text{grav}} = 6.6743\text{e-}11 \times 1.989\text{e}41 / (5\text{e}20)^2 = 5.311\text{e-}11 \text{ m/s}^2$$

#### Step 2: Cosmic Expansion Factor

$$H(z) = 2.269\text{e-}18 \text{ s}^{-1}; H(z) \times t = 2.269\text{e-}18 \times 1.578\text{e}17 = 0.358; \text{factor} = 1.358$$

#### Step 3: SFR Mass Fraction

$$M_{\text{sf}} = 0.045; 1 + M_{\text{sf}} = 1.045$$

#### Step 4: Time-Reversal Correction

$$f_{\text{TRZ}} = 0.04; 1 + f_{\text{TRZ}} = 1.04$$

#### Step 5: Gravitational Total

$$g_{\text{grav\_total}} = 5.311\text{e-}11 \times 1.358 \times 1.045 \times 1.04 = 7.856\text{e-}11 \text{ m/s}^2$$

#### Step 6: Aether EM Correction

$$a_{\text{EM}} = (1.602\text{e-}19 \times 1\text{e}5 \times 1\text{e-}5 / 1.673\text{e-}27) \times 11 \times 1\text{e-}12 = 1.053\text{e-}3 \text{ m/s}^2$$

#### Step 7: Final Solution

$$g_{\text{M74}} = 7.856\text{e-}11 + 1.053\text{e-}3 \approx 1.053\text{e-}3 \text{ m/s}^2$$

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### 4. Physical Interpretation

M74's near-perfect symmetry and moderate SFR place it exactly in the standard UQFF Sbc spiral category. JWST MIRI imagery showing crisp HII regions distributed along symmetric arms confirms that M74 is in a quiescent, undisturbed star-formation mode. The consistent  $g_{\text{M74}} = 1.053 \times 10^{-3} \text{ m/s}^2$  result joins NGC 2841 (PAPER\_775), NGC 6217 (PAPER\_777), and NGC 7049 (PAPER\_779) in affirming UQFF universality across nearby spiral morphologies.

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### 5. Conclusions

UQFF applied to M74 Phantom Galaxy yields  $g \approx 1.053 \times 10^{-3} \text{ m/s}^2$ . As JWST's 2022 MIRI showcase target, M74 provides a clean observational anchor for standard UQFF Sbc galaxy calculations at  $z = 0.0022$ .

*PAPER\_781, CP4 class #365. v5.42.*

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                        | SM / Experiment                        | Source                              | Alignment    |
|----------------------------------|------------------------------------------------------------------------|----------------------------------------|-------------------------------------|--------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                       | 1/137.036                              | PDG 2024                            | ✓ Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)                | $1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018                         | ✓ Consistent |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$     | Super-K 2024                        | ✓ Consistent |
| UQFF buoyancy signature          | F_U_Bi_i unique gravitational correction                               | Not yet measured                       | Future gravitational wave detectors | Testable     |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F\_U\_Bi\_i) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*